Fiscal Year:	FY 2008	**Task Last Updated:**	FY 06/11/2008
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Measurement of Blood and Tissue Chemistry		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	babs.soller@reflectancemedical.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	508-366-4700
Organization Name:	Reflectance Medical Inc.		
PI Address 1:	116 Flanders Road, Suite 1000		
PI Address 2:			
PI Web Page:			
City:	Westboro	**State:**	MA
Zip Code:	01581	**Congressional District:**	3
Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	10/01/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: changed end date to 10/1/2007 to accommodate FY08 NSBRI submission for final report (jp 5/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heard, Stephen (University of Massachusetts Medical School)		
Grant/Contract No.:	NCC 9-58-SMS00403		
Performance Goal No.:			
Performance Goal Text:			

Trauma and acute medical problems along with loss of skeletal muscle mass, strength and endurance, are some of the most serious risks facing astronauts during long duration spaceflight. The measurement of two metabolic parameters, muscle pH and oxygen can be applied in both areas. This project examines the hypothesis that near infrared spectroscopy (NIRS) is a platform technology that can provide noninvasive physiologic monitoring in support of multiple NASA needs. The immediate goal is to produce, validate and deliver to NASA-JSC a small, lightweight medical monitor which utilizes NIRS to measure important metabolic parameters. Requisite precision and accuracy will be demonstrated for both male and female subjects of any ethnic origin. The device will perform equally well on the forearm and the thigh, despite gender and weight related differences in fat thickness. The specific aims of this project are: (1) Develop calibration procedures for modified NIRS sensor to measure deep muscle metabolic parameters (tissue pH and oxygen); (2) Validate sensor in exercise protocol and deliver systems to NASA-JSC; (3) Determine values for tissue pH and oxygen that can be used by a smart medical system to indicate shock and assist in guiding treatment; and (4) Evaluate hardware for flight requirements and develop plan to produce flight-ready instrumentation.
Over the course of this project we improved our understanding of how light interacts with muscle covered by thick layers of fat. We developed a mathematical model of spectra for human tissue which takes into account the skin, fat and muscle layers. The model was used, in conjunction with tissue mimicking phantoms, to better understand the influence that fat thickness and its absorbance and scattering properties have on the absorption spectrum used to calculate muscle pH and PO2. We created a novel tissue mimicking phantom based on a near infrared dye with absorption properties similar to deoxyhemoglobin. Measurement on these phantoms was used to demonstrate prediction capabilities of the Monte Carlo model used to study light interaction with multi-layer tissue structures. Using these phantom materials we developed and demonstrated new methods for correcting tissue absorption spectra for skin pigment and fat using a novel 2-source fiber optic sensor and mathematics to allow us to isolate the spectrum of the muscle from the spectrum of the skin and fat.

The phantoms and Monte Carlo model helped us identify key sensor design issues to assure optical depth penetration through fat on the forearm and the thigh. One system was optimized for thinner fat layers and was made available to NASA-JSC for handgrip studies and measurements on the calf during treadmill exercise. We redesigned this sensor and the accompanying monitor to achieve good muscle spectra through thick layers of fat overlying thigh muscle. One of these systems was delivered to NASA-JSC for thigh measurements during treadmill studies.

Task Description:

A new method for calculating muscle oxygen saturation (SmO2) was developed and its precision was determined as part of a handgrip study at NASA-JSC. We also work with NASA-JSC to demonstrate that we could use the pH equation to noninvasively determine hydrogen ion threshold as a surrogate for lactate threshold during cycle ergometry.

Another goal of our project was to develop noninvasive methods to measure absolute values for key parameters so they can be used as part of a smart medical system to help diagnose and guide treatment for critical injuries. To do this we needed to establish normal values for new parameters and be able to identify values of these new parameters that indicate when someone is sick or getting better. Working with the US Army we used lower body negative pressure (LBNP) as a model to simulate the early stages of hemorrhagic shock or internal bleeding. We demonstrated specific values for SmO2 and PmO2 as early indicators of internal bleeding and showed that they provided significantly earlier warning than currently used clinical parameters.

We also conducted a study on sepsis patients undergoing resuscitation with the standard clinical protocol “Early Goal Directed Therapy” (EGDT) and showed that SmO2 could be used to indicate when a patient was under-resuscitated while noninvasively determined muscle pH indicated when patients were over-resuscitated with chloride-containing solutions which cause acidosis. Specific values for these parameters were established to provide noninvasively determined goals to direct treatement of septic patients.

We have developed and demonstrated accurate methods for determining muscle oxygen and pH for sick and healthy individuals independent of their skin color and fat thickness. We have demonstrated that these methods can be used on exercising individuals and have applicability for very sick patients being treated in the emergency room. These advances position us to apply a combination of these parameters for determining oxygen consumption during EVA and assessing muscle and aerobic deconditioning during space exploration.

The NIRS noninvasive metabolic monitor is expected to have many applications for NASA. The system will have additional use on earth for military and civilian personnel treating critically ill and injured patients. It can also be used in the hospital, ambulances and helicopters. As part of a Smart Medical System, advanced medical assessment and monitoring may become available to physicians in remote and rural areas, who may not have access to specialist expertise.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This work will have direct earth based application. The prototype monitors we have developed will have application in emergency response vehicles, emergency rooms, and hospitals. Pre-hospital applications include assessing the severity of shock and triaging multiple casualties, as well as providing a sensor for a smart medical system to guide resuscitation from hemorrhage. In the ICU we expect that this monitor will find application in distinguishing between hemorrhagic and septic shock and helping to assess the effectiveness of sepsis treatments. The direct muscle application of interest to NASA for assessing fitness in space, may be useful to assess success of physical therapy in rehabilitating patients with muscle injury or atrophy. There is also general medical application for the diagnosis of anemia and if small and inexpensive enough, screening world-wide for malnutrition. There is also a possible application for the diagnosis of diabetic foot ulcers. A smaller version of this monitor could find use in the training of elite and weekend athletes.

1) Calibration Equations
We developed a novel method to correct transdermal spectra for interference from overlying layers of skin and fat. This method employs a specially designed sensor and novel mathematics. This methodology was published and is patent pending. We then found that we also needed to correct for variation in muscle optical properties between subjects. We developed a new mathematical method for this and demonstrated it in a phantom study, and on pH measurements for exercising subjects. A paper on this technique has recently been published in the journal Applied Spectroscopy. A patent application has been filed.

We developed a new method for determining muscle oxygen saturation (SmO2) A paper has on has been published in

Task Progress:	the journal Optics Express and a patent application filed. We independently validated the pH measurement by showing that anaerobic threshold calculated from hydrogen ion concentration (derived from pH) was highly correlated with the lactate threshold during a VO2max test. This paper is accepted for the Journal of Applied Physiology. A patent is pending on this technology. 2) Leg System A system for use on the leg during cycling was completed and delivered to NASA-JSC Exercise Physiology Lab. The leg system was tested on 10 subjects doing an VO2max cycling test. A special sensor holder was designed to hold the sensor against the skin and completely eliminated motion artifacts from spectra during cycling, walking and running. The system has been used to collect data during several treadmill tests. The first test was in the exercise physiology lab (N=10) and 2 sensors were simultaneously used, one on the calf and one on the thigh to look at the differences in SmO2 between the 2 muscles and how they contribute to whole body VO2. We also collected data from the thigh and calf during the shirtsleeve portion of the EVA Program's Suit Test 1 and Suit Test 2 protocols in the POGO simulator. This allowed us to collect data during walking and running under simulated partial gravity conditions. Data analysis is on-going and will continue as part of the renewed project. 3) Trauma Care We completed enough subjects to demonstrate that in patients with severe sepsis SmO2 was highly correlated with blood lactate. This very important finding provides a noninvasive method for assessing severity of illness and adequacy of resuscitation. SmO2 is an indicator of microvascular perfusion and low SmO2 indicates very poor blood flow, inadequate oxygen delivery, and consequently production of lactate. We determined on a preliminary basis that SmO2 of 40% corresponds to lactate of 4 mmol/l and suggest that a treatment goal might be to resuscitate to assure that SmO2 is significantly greater than 40%. With the US Army we demonstrated that SmO2 and PmO2 were very early indicators of internal bleeding, highly correlated with stroke volume reduction and much earlier than changes in HR, BP and pulse oximetry.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Abstracts for Journals and Proceedings	Marengi N, Yang Y, Lee SM, Wilson C, Soller BR. "Assessment of metabolic rate in a spacesuit: measuring oxygen consumption without a facemask." Aerospace Medical Association 79th Annual Meeting, Boston, MA, May 11-15, 2008. Aviat Space Environ Med 2008 Mar;79(3):L294. , Mar-2008
Abstracts for Journals and Proceedings	Soller BR, Soyemi O, Landry M, Shear M, Wu J, Hagan RD. "Validation of a new NIRS method for measuring muscle oxygenation during rhythmic handgrip exercise." American College of Sports Medicine 53rd Annual Meeting, Denver, Co., May 30 - June 3, 2006. Med Sci Sports Exerc. 2006;38:S248. , May-2006
Abstracts for Journals and Proceedings	Soller BR, Yang Y, Lee SMC, Wilson CA. "Comparison of noninvasively determined hydrogen ion and lactate thresholds during cycle exercise." American College of Sports Medicine 54th Annual Meeting , New Orleans, LA., May 30 - June 2, 2007. American College of Sports Medicine 54th Annual Meeting , May 30 - June 2, 2007. Program and abstracts. , May-2007
Articles in Peer-reviewed Journals	Soller BR, Hagan RD, Shear M, Walz JM, Landry M, Anunciacion D, Orquiola A, Heard SO. "Comparison of intramuscular and venous blood pH, PCO(2) and PO(2) during rhythmic handgrip exercise." Physiol Meas. 2007 Jun;28(6):639-49. PMID: 17664618 , Jun-2007
Articles in Peer-reviewed Journals	Soller BR, Yang Y, Lee S, Wilson C, Hagan RD. "Noninvasive determination of exercise-induced hydrogen ion threshold through direct optical measurement." J Appl Physiol. In press, September 2007. , Sep-2007
Articles in Peer-reviewed Journals	Soller BR, Yang Y, Soyemi OO, Ryan KL, Rickards CA, Walz JM, Heard SO, Convertino VA. "Noninvasively determined muscle oxygen saturation is an early indicator of central hypovolemia in humans." J Appl Physiol. In press, 2007. Epub 2007 Nov 15. PMID: 18006869 , Nov-2007
Articles in Peer-reviewed Journals	Yang Y, Soyemi O, Scott PJ, Landry MR, Lee SM, Stroud L, Soller BR. "Quantitative measurement of muscle oxygen saturation without influence from skin and fat using continuous-wave near infrared spectroscopy." Optics Express. 2007 Oct;15(21):13715-30. http://dx.doi.org/10.1364/OE.15.013715 , Oct-2007
Patents	US Patent 7,245,373. Patent, July 2007. Jul-2007 Phillipps P, Soller BR, Parker M. "Spectroscopic System for Reflectance Measurements."
Patents	Patent Application 60/949,789. Patent Application, July 2007. Jul-2007 Soller BR. "Physical Performance Monitoring and Monitors."
Patents	US Application 11/411,538 and PCT/US2006/015955. Patent, April 2006. Apr-2006 Yang Y, Soller BR, Soyemi O, Shear M. "Systems and Methods for Correcting Optical Reflectance Measurements."
Patents	Application 11/755,643. Patent application, May 2007. May-2007 Yang Y, Soyemi OO, Soller BR. "Measuring Tissue Oxygen Saturation."